

Karuniya Premnath

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EDUCATION

Northeastern University , Boston, MA	Jan 2026 – Present
Master of Science in Data Science	Expected May 2028
Relevant Coursework: Machine Learning, Essentials of Data Science, Machine Learning Operations	CGPA: 3.667
Anna University , Tiruchirappalli, Tamil Nadu, India	Nov 2021 – Apr 2025
Bachelor of Technology in Artificial Intelligence and Data Science	
Relevant Coursework: Data Science, Machine Learning, Deep Learning, Artificial Intelligence	

SKILLS

Programming: Python, SQL, R
Machine Learning: SVM, LSTM, TensorFlow, Keras, Scikit-learn
Data Visualization: Matplotlib, Seaborn
Tools & Concepts: Google Maps API, Audacity, Deep Learning, IoT Integration, Predictive Analytics
Certifications: Executive PG Certification in Data Science & AI – IIT Roorkee (Intellipaat); Advance SoC Design – NIT Trichy (PROBE 2023); Cloud Solutions Workshop – Zoho Catalyst

WORK EXPERIENCE

Gnani.ai Software Developer Intern Bengaluru, India	July 2024
– Enhanced speech clarity using Audacity through advanced noise reduction and volume normalization	
– Supported AI-based speech processing workflows during live project implementation	
– Improved presentation-quality audio outputs for deployment-ready systems	
Infilabs Software Developer Intern Tiruchirappalli, India	Aug 2023
– Designed and optimized data structures for efficient storage and retrieval across AI-driven modules	
– Built and tested data-driven modules, ensuring clean, well-structured datasets for reliable processing	
– Collaborated with the development team on debugging, code reviews, and deployment support	

PROJECTS

Youth Tobacco Use Prediction and Policy Analysis (GYTS Multi-Country Study)	April 2026
– Lacked data-driven insight into which behaviors and policies most drive youth smoking across 187 countries.	
– Integrated GYTS, WHO MPOWER, and World Bank data into a 522-country-year panel; trained a regularized XGBoost model with SHAP explainability and K-Means/UMAP clustering.	
– Identified peer exposure and secondhand smoke as top predictors and showed policy upgrades could cut prevalence by up to 7.34 percentage points in high-risk countries.	
A Data Visualization-driven Decision Support System for Smart Agriculture	April 2026
– Agricultural AI tools focused on model accuracy while leaving farmers unable to interpret raw outputs, limiting real-world usability for non-technical smallholders.	
– Built an Android app integrating a CNN for disease detection, XGBoost for crop recommendation (2,200 records, 22 crops), and a soil analysis module; used Firebase for real-time sync and designed color-coded, farmer-readable visualizations for every AI output.	
– Delivered a unified visualization-first platform uncovering key agronomic patterns (P–K correlation $r=0.74$, crop-specific NPK clustering), making AI-driven recommendations immediately actionable for low-literacy, non-technical users.	
Wireless Cylinder Safety Monitoring System	August 2025
– Situation: Gas leakage in industrial and domestic environments posed safety risks due to the absence of continuous real-time monitoring and automated emergency alerting.	
– Action: Designed an IoT-based gas leakage detection system with wireless data transmission for continuous safety tracking; developed an alert notification system for emergency response.	
– Result: Delivered a functional real-time monitoring system enabling immediate emergency response through automated alerts and continuous wireless safety tracking.	